

TUKA CURIOUS MINDS [TCM] ROBO RUMBLE

BUDGET ESTIMATES + BOM

Planned Exchange Rate Baseline: 1.00 USD = 18.00 ZAR

Component & Logistics Grid

Category	Component / Expense Description	Qty	Cost (USD)	Cost (ZAR / R)	Resource Status / Supplier
Chassis	goBILDA 1120/2203 Framework Channels	—	\$0.00	-	On Hand (Kit Inventory)
Chassis	Metric Fasteners (M3/M4 Screws & Locknuts)	1 pk	\$0.00	-	On Hand (Kit Inventory)
Motion	REV Core Hex Propulsion Motors (125 RPM)	4	\$0.00	-	On Hand (Kit Inventory)
Motion	goBILDA Traction & Passive Omni Wheels	4	\$0.00	-	On Hand (Kit Inventory)
Weapon	REV 6,000 RPM High-Speed Motor	1	\$0.00	-	On Hand (Kit Inventory)
Weapon	12mm Hardened Steel Cross Dead-Shaft Axis	1	\$21.00	378,00	McMaster-Carr (6112K21)
Weapon	HTD 5mm Flexible Timing Belt (15mm Wide)	2	\$25.00	450,00	Misumi USA (HTD5-15-400)
Actuation	12V 25N Push-Pull Linear Solenoids	2	\$42.00	756,00	Micro Robotics (JF-1039B)
Electronics	REV Control Hub MCU Main Logic Board	1	\$0.00	-	On Hand (Kit Inventory)
Electronics	12V 3.6Ah NiMH Heavy-Duty Power Cell	1	\$0.00	-	On Hand (Kit Inventory)
Electronics	12V Single-Channel Isolated Optical Relay	1	\$3.50	63,00	DigiKey (2448-1014-ND)
Safety	High-Visibility Red Twist-Lock E-Stop Button	1	\$18.95	341,10	RS Components (764-4325)

Safety	High-Current Toggle Primary Power Switch	1	\$9.50	171,00	DigiKey (360-2092-ND)
Safety	20A Inline ATM Mini Blade Fuse Module	1 set	\$4.50	81,00	AutoZone (BP/HHA-RP)
Armor	2mm Mild Steel Armor Face Plate Offcuts	—	Scrap	-	On Hand (Free Workshop Scrap)
Armor	1.5mm Aluminum Casing Shell Sheets	—	Scrap	-	On Hand (Free Workshop Scrap)
Tools	1mm Thin Inox Grinder Cutting Wheels	1 pk	\$20.00	360,00	Local Hardware Consumables
Build	Components, Tools			R 2 600,10	
Operations	Travel Costs: Daveyton to Midrand x 2	—	—	R 2 500,00	Transport Allocation
Operations	Sustenance: Meals & Refreshments	—	—	R 840,00	R70pp x 6 Crew x 2 Days
Operations	Custom Team T-Shirts (In-House Heat Press)	6	—	R 900,00	Blanks (R600) + Ink (R150) + Del (R150)
				R 6 840,10	

SUMMARY A: COMPONENT & PARTICIPATION SUB-TOTAL

Procured Robot Hardware Components: R2,240.10 (*Solenoids, static shaft axis, timing belts, optical relay, and electrical safety switches*)
On-Hand Kit Assets Value: R0.00 (*Zero out-of-pocket tracking by fully maximizing existing inventory components*)
Chassis Fabrication Consumables: R360.00 (*Grinder cutting wheels*)
SUB-TOTAL COMPONENT VALUE: R2,600.10

HEADING SUMMARY B: TEAM LOGISTICS & OPERATIONAL SUB-TOTAL

Team Site Transit Fees: R2,500.00 (*Daveyton to Midrand x 2 sessions*)
Sustenance & Refreshments: R840.00 (*Budgeted localized safety ceiling tracking at R70 per person*)
Custom Corporate Identity Apparel: R900.00 (*Blank fabric shells, sublimation ink base lines, and cross-district delivery loops utilizing y*)
SUB-TOTAL OPERATIONAL LOGISTICS: R4,240.00

OVERALL PROJECT IMPLEMENTATION TOTAL

COMBINED SYSTEM VALUE ENVELOPE: R6,840.10

(Calculated cleanly as: R2,960.10 Structural Hardware Component Investment + R4,240.00 Operational Logistics)

es)

our on-hand machinery)